

DisasterSAGE: Evidence-to-Action Learning for Disaster Response from Remote Sensing

Anonymous submission

Abstract

Vision-language models (VLMs) have shown promising capabilities in understanding remote sensing imagery, but disaster response requires more than visual recognition or question answering. Turning complete visual evidence into verifiable, rule-constrained decisions remains difficult because current supervision rarely records how evidence is grounded, computed, and converted into actions. To address this gap, we introduce **DisasterSAGE**, an Evidence-to-Action (E2A) framework for disaster reasoning. **DisasterSAGE-Agent** first constructs verified E2A traces by integrating mask-complete evidence, documented professional knowledge, and deterministic execution. From these traces, we build **DisasterSAGE-Bench**, which provides 75,843 image-conditioned tasks from 42 real-world events and 13 terrestrial and marine disaster types, with 245 executable templates spanning 5 levels from area and count to optimization and cost. The benchmark requires models to preserve visual grounding while performing rule-based numerical, spatial, graph, and optimization operations. We therefore develop **Disaster-Codex**, a 7B vision-code model initialized through task-vector merging and aligned with E2A supervised fine-tuning, Grounding reinforcement learning, and Action reinforcement learning. Experiments reveal substantial limitations of open-source VLMs on higher-level E2A tasks. On the fixed test split, Disaster-Codex improves the five-level macro score from 10.79 to 66.41 and the disaster-type macro score from 19.61 to 67.80.

Introduction

Rapidly assessing damaged targets, infrastructure conditions, and potential risks is fundamental to rescue resource allocation and recovery planning. Remote sensing imagery provides broad-area, timely, and repeatable disaster observations (Kaku 2019), while vision-language models (VLMs) make these observations queryable and interpretable in natural language. Recent remote sensing VLMs support instruction following in RSGPT (Hu et al. 2025), multisensor image comprehension in EarthGPT (Zhang et al. 2024a), and grounded dialogue in GeoChat (Kuckreja et al. 2024). Yet response planning often requires literature-grounded operational assessments: which structures fall within a wildfire exposure-distance threshold (Nicoletta et al. 2023), how wildfire exposure changes road-network connectivity (Arango et al. 2023), how road-network recovery time varies with damage level and available recovery re-

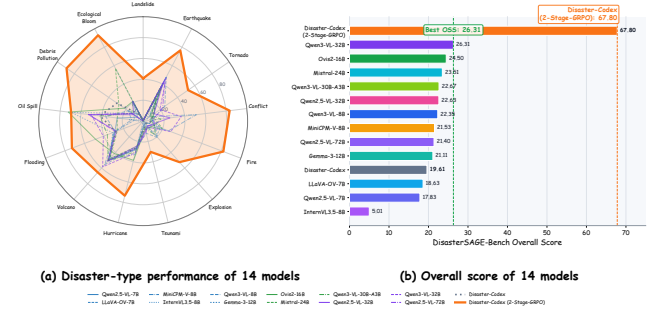


Figure 1: DisasterSAGE-Bench result preview: (a) disaster-type macro scores and (b) overall ranking of representative open VLMs and staged Disaster-Codex checkpoints, including the final two-stage GRPO model.

sources (Arango et al. 2025), and how debris clearance and equipment repositioning can restore supply-demand connectivity cost-effectively (Sayarshad, Du, and Gao 2020). These assessments require task-relevant visual evidence to be recovered, converted into computable states, and evaluated with source-grounded spatial, graph, temporal, and optimization rules.

Learning this evidence-to-action process requires supervision of intermediate states and rule-governed operations, yet existing datasets cover only parts of it: xBD (Gupta et al. 2019) covers building damage, FloodNet (Rahnemoonfar et al. 2021) supports post-flood understanding and question answering, and DisasterM3 (Wang et al. 2025a) provides image-language supervision across hazards, sensors, and tasks. Most datasets in Table 1 supervise endpoints—labels, masks, questions, or final text—rather than an explicit executable transition from evidence to rule-evaluated states. This endpoint-centered supervision limits the auditability of answer generation. A correct answer may still be produced from incomplete evidence or incorrect evidence-variable binding, while an incorrect answer provides little indication of whether the failure arises from grounding, state construction, calculation, spatial reasoning, or action mapping.

This ambiguity is especially problematic for disaster assessment, where actionable outputs require a complete path from evidence to response: evidence must be grounded,

Dataset	Data Modality		Scope		Annotation and Tasks						Scale		
	Opt.	SAR	MH	Mar.	Perc.	Text	Grd.	Quant.	Spatial Reason.	Traj.	Haz.	Events	Text Pairs
xBD	✓	✗	✓	✗	✓	✗	✗	✗	✗	✗	6	19	–
Sen1Floods11	✓	✓	✗	✗	✓	✗	✗	✗	✗	✗	1	11	–
Incidents1M*	✓	✗	✓	✗	✓	✗	✗	✗	✗	✗	43 [†]	–	–
FloodNet	✓	✗	✗	✗	✓	✓	✗	✓	✗	✗	1	1	11K
SpaceNet 8	✓	✗	✗	✗	✓	✗	✗	✗	✓	✗	1	2	–
RescueNet	✓	✗	✗	✗	✓	✗	✗	✗	✗	✗	1	1	–
MADOS [§]	✓	✗	✗	✓	✓	✗	✗	✗	✗	✗	2 [‡]	–	–
BRIGHT	✓	✓	✓	✗	✓	✗	✗	✗	✗	✗	7	14	–
LADI v2	✓	✗	✓	✗	✓	✗	✗	✗	✗	✗	–	>50	–
DisasterM3	✓	✓	✓	✗	✓	✓	✓	✓	✓	✗	10	36	123K
DisasterVQA*	✓	✗	✓	✗	✗	✓	✗	✓	✗	✗	–	–	4.4K
DisasterSAGE-Bench	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	13	42	75.8K

[†] Incident labels rather than disaster types. [‡] Primary pollution targets (marine debris and oil spill), alongside 15 semantic classes. [§] 174 Sentinel-2 scenes rather than distinct disaster events. * Ground-level social-media imagery, not remote sensing imagery.

Table 1: Comparison of representative visual disaster datasets by data modality, scope, supervision, task coverage, and scale. *Abbreviations:* Opt., optical or RGB imagery; SAR, synthetic aperture radar imagery; MH, multi-hazard scope; Mar., marine coverage; Perc., perception labels; Text, textual or language annotations; Grd., language-conditioned grounding; Quant., numerical tasks; Spatial Reason., relational or network reasoning; Traj., structured trajectories; Haz., number of disaster types. Events and Text Pairs denote distinct historical disaster events and image-conditioned prompt–target pairs, respectively. ✓ and ✗ indicate supported and unsupported attributes.

bound to scene states, evaluated with documented rules, and converted into reproducible actions. We define this path as the *Evidence-to-Action (E2A) trace*, comprising evidence, observation, state, knowledge, impact, and action. Existing evidence-seeking methods address only part of this requirement. GeoReason (Li et al. 2026) aligns reasoning traces with final decisions, and RS-EoT (Shao et al. 2026) supports evidence-grounded reasoning through iterative visual evidence seeking. However, these methods do not provide complete E2A traces as scalable supervision.

To provide such supervision, we propose DisasterSAGE-Agent, an offline trace-generation agent that integrates remote sensing imagery, complete object masks, executable task templates, professional references, and deterministic program outputs into six-field E2A traces. After generation, automatic verification checks visual-evidence completeness, object counts, coordinates, finite values, units, schema validity, and consistency under re-execution. Only traces that pass these checks are retained as auditable process-level supervision.

Based on the verified traces, we establish DisasterSAGE-Bench with 75,843 image-conditioned tasks generated from 245 executable templates. These tasks cover 42 real-world events and 13 terrestrial and marine disaster types. The templates define 5 levels of operational complexity: L1 area and count, L2 compactness and shape, L3 proximity and buffer, L4 graph and connectivity, and L5 optimization and cost. Rather than assigning only an endpoint label, each instance pairs complete visual evidence with a verified trajectory and a deterministic target, enabling both final-answer evaluation and diagnosis of grounding or action-execution errors. The targets are reproducible under encoded rules, rather than intended as universal decisions for field response. Thus each instance supervises the path to an answer, not merely the an-

swer itself.

DisasterSAGE-Bench couples remote sensing grounding with numerical, spatial, graph, and optimization reasoning. Its executable templates require models to bind visual evidence to variables, manipulate formulas, track intermediate states, and execute structured operations. These requirements make code-model priors particularly suitable for higher-level E2A tasks. Therefore, we develop Disaster-Codex, a 7B vision–code model initialized through compatible task-vector merging. Its training-free initialization preserves the full visual pathway while injecting code-model priors into the shared language backbone, enabling a single image-conditioned model to combine visual grounding with latent programmatic reasoning without an external code engine at inference. E2A supervised fine-tuning (E2A SFT) establishes the trace protocol, Grounding reinforcement learning (Grounding RL) improves complete evidence recovery, and Action reinforcement learning (Action RL) aligns rule-constrained outputs with answer semantics and format. Figure 1 previews the resulting empirical gap: representative open VLMs remain far below staged Disaster-Codex checkpoints on the fixed test split, especially at higher E2A levels.

Our contributions are threefold:

- We introduce DisasterSAGE-Agent to generate verified E2A traces from complete evidence, professional sources, and deterministic execution.
- We build DisasterSAGE-Bench with 75,843 tasks from 42 events, 13 disaster types, and 245 templates across five reasoning levels.
- We develop Disaster-Codex, a 7B image-conditioned vision–code model that couples compatible task-vector initialization with E2A SFT, Grounding RL, and Action RL for verifiable evidence-to-action reasoning.

Related Work

Remote sensing vision–language reasoning. Existing research covers visual question answering with RSVQA (Lobry et al. 2020) and EarthVQA (Wang et al. 2024); contrastive image–text pretraining with RemoteCLIP (Liu et al. 2024) and RS5M/GeoRSCLIP (Zhang et al. 2024b); instruction-following models such as RSGPT (Hu et al. 2025), EarthGPT (Zhang et al. 2024a), and GeoChat (Kuckreja et al. 2024); and the multitask benchmark VRS-Bench (Li, Ding, and Elhoseiny 2024). A recent survey (Li et al. 2024b) provides a broader overview of the field. These works make remote sensing imagery more accessible through question answering, retrieval, captioning, and dialogue, but their supervision is usually attached to answers or language responses rather than to executable intermediate states. Beyond perception-oriented tasks, GeoReason (Li et al. 2026) improves consistency between reasoning traces and final decisions, while RS-EoT (Shao et al. 2026) promotes evidence-grounded reasoning through iterative visual evidence seeking. RescueADI (Liu et al. 2025) further plans and executes tools for perception, counting, area calculation, and path finding. Nevertheless, these methods do not explicitly connect visual evidence to documented disaster variables and executable decision rules.

Remote Sensing Datasets for Disaster Vision Tasks. Visual disaster datasets mainly supervise perception or language endpoints. xBD (Gupta et al. 2019), BRIGHT (Chen et al. 2025), and LADI v2 (Scheele, Picchione, and Liu 2025) cover damage; Sen1Floods11 (Bonafilia et al. 2020), FloodNet (Rahnemoonfar et al. 2021), SpaceNet 8 (Hänsch et al. 2022), and RescueNet (Rahnemoonfar, Chowdhury, and Murphy 2023) cover water, roads, buildings, or UAV scenes. Incidents1M (Weber et al. 2023) and Disaster-VQA (Al-Mohannadi et al. 2026) use social-media imagery for recognition and VQA; MADOS (Kikaki et al. 2024) segments marine pollutants, and DisasterM3 (Wang et al. 2025a) provides multi-hazard remote sensing image–language supervision. DisasterInsight (Tehrani et al. 2026) evaluates building function, damage, counting, and grounded reports. However, these datasets do not expose a complete executable trace from evidence and domain knowledge to rule-evaluated states and responses.

Model merging and programmatic reasoning. Model merging combines compatible models in parameter space to integrate complementary capabilities without training a new model from scratch. In multimodal settings, it can transfer capabilities such as coding while preserving the visual pathway. Model soups (Wortsman et al. 2022) and TIES-Merging (Yadav et al. 2023) use parameter averaging and interference reduction, while VisCodex (Jiang et al. 2026) and REMEDY (Zhu et al. 2025) extend merging to multimodal models. Program-of-thought prompting (Chen et al. 2023) and PAL (Gao et al. 2023) instead execute generated programs with an external interpreter. However, these methods do not jointly preserve fine-grained disaster evidence and align it with formula, spatial, graph, and optimization operations in a single image-conditioned model.

DisasterSAGE

We propose DisasterSAGE-Agent for verified E2A trace generation and use the resulting traces to construct DisasterSAGE-Bench. This section first describes how the agent combines annotations, disaster references, and deterministic execution, and then explains how the traces form benchmark instances across events and hazards.

DisasterSAGE-Agent

Each task is generated from an executable template τ with four components: required object classes $\mathcal{C}$, ground sample distance g , deterministic operation F , and target serializer H . We write this template as $\tau = (\mathcal{C}, g, F, H)$. The required-class masks $M_{\mathcal{C}}$ and the mask-to-box operator $\mathcal{B}$ then produce mask-complete evidence E_{τ} , executed state v_{τ} , and target y_{τ} :

$$\begin{aligned} E_{\tau} &= \mathcal{B}(M_{\mathcal{C}}), \\ v_{\tau} &= F(M_{\mathcal{C}}; g), \\ y_{\tau} &= H(v_{\tau}). \end{aligned} \tag{1}$$

Eq. (1) links evidence, state, and target under literature-specified variables, thresholds, operators, quantities, and costs. Each verified trace is a six-field record $z = (e, o, s, k, i, a)$: evidence e ; observation o ; state s ; knowledge k ; impact i ; action a . The trace state s serializes the executed state v_{τ} . Four model-backed roles process the trace sequentially.

Evidence–Observer records required objects and image-supported observations from the image, query, and mask-derived object set (Qwen2.5-VL-72B-Instruct (Bai et al. 2025b)). **State–Knowledge** binds objects to template variables and executed values, then records the retrieved formula, threshold, or rule (DeepSeek-R1-Distill-Llama-70B (Guo et al. 2025)). **Impact–Action** expresses the verified state’s operational meaning and rule-constrained procedure (Llama-3.3-70B-Instruct (Dubey et al. 2024)). **Trace Verifier** reviews the trace against the task specification, reference material, and deterministic target (Skywork-Critic-Llama-3.1-70B (Tu et al. 2024)). Each field pair is passed to the next role.

These models organize authoritative inputs rather than determine task truth. Acceptance requires deterministic checks of evidence completeness, coordinates, template variables, finite values, units, schema closure, and agreement with re-execution. Failed instances are regenerated or removed, yielding auditable supervision for encoded procedures rather than unconstrained rationales.

DisasterSAGE-Bench

We construct the image database from DisasterM3 and MADOS. Using DisasterSAGE-Agent to generate verified E2A traces, we build DisasterSAGE-Bench as a multi-hazard executable benchmark (Figure 2). As summarized in Table 2, DisasterSAGE-Bench contains 75,843 image-conditioned instances from 42 real events across five continents and 13 terrestrial and marine hazards. Each stores the image, question, complete evidence, trace, and target; states and targets follow documented rules, making the benchmark re-

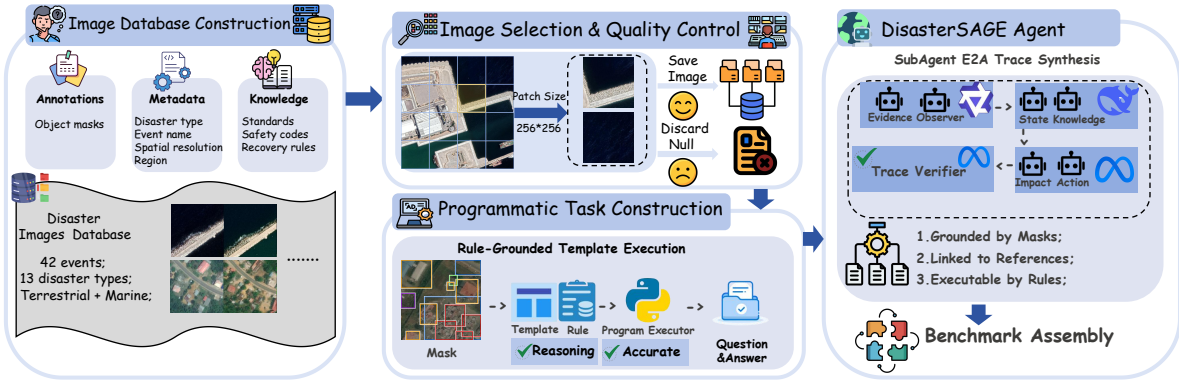


Figure 2: Construction of DisasterSAGE-Bench with DisasterSAGE-Agent, covering image database assembly, quality control, programmatic task generation, and verified E2A trace synthesis.

executable rather than a collection of field-response decisions.

Its 245 templates form five E2A levels (Figure 3): L1 area/count, L2 compactness/shape, L3 proximity/buffer, L4 graph/connectivity, and L5 optimization/cost. Categorical targets retain identifier and semantic interval, binary targets explicit meaning, and quantitative targets value and unit. Terrestrial and marine scenes use 0.8 and 10 m per pixel, respectively. Complete masks yield 457,339 boxes. The fixed split contains 60,669 training, 7,587 validation, and 7,587 test instances. All stages use the same split, enabling consistent comparison across model checkpoints.

Raster graph execution. Road masks are thinned to one-pixel centerlines and converted into an eight-neighbor graph; building centroids snap to the nearest centerline, and the largest component defines the serviceable core. Obstruction-aware tasks remove affected pixels and invalid diagonal transitions. Routing weights orthogonal and diagonal steps by g and $\sqrt{2}g$ and applies multi-source Dijkstra with template-specific costs. Empty graphs and invalid denominators are rejected. These operators generate targets only; masks and graphs remain hidden at evaluation.

Disaster-Codex

DisasterSAGE-Bench contains image-conditioned tasks coupling remote sensing grounding with numerical, spatial, graph, and optimization operations. They require evidence recovery and code-like reasoning over formulas, variables, graphs, and programs. We construct a 7B Disaster-Codex by transferring a code-task direction into a VLM-compatible language backbone while retaining visual modules.

For training-free initialization, we merge architecture-compatible language parameters around a shared base θ_b while preserving the VLM visual pathway. The vision-language checkpoint θ_v and code-language checkpoint θ_c define task directions weighted by λ . Using established weight-space composition (Wortsman et al. 2022) under vision-code compatibility constraints (Jiang et al. 2026), the

Statistic	Value
Task diversity	
Task/action templates	245
Disaster events / represented continents	42 / 5
Spatial-resolution regimes	0.8–10 m per pixel
E2A trace supervision	
Average reasoning question length	274.84
Average reasoning process length	1,234.28
Maximum reasoning question length	548
Maximum reasoning process length	2,631
Evaluation protocol	
Train / validation / test	60,669 / 7,587 / 7,587

Table 2: DisasterSAGE-Bench construction statistics. The reasoning process is the concatenated E2A trace from evidence to action, excluding the final answer. Lengths are computed from raw serialized strings before tokenization.

directions Δ_v, Δ_c and Disaster-Codex initialization θ_{DC} are

$$\begin{aligned} \Delta_v &:= \theta_v - \theta_b, & \Delta_c &:= \theta_c - \theta_b, \\ \theta_{DC} &= \theta_b + \lambda \Delta_v + (1 - \lambda) \Delta_c. \end{aligned} \quad (2)$$

Eq. (2) merges only language parameters and retains all VLM visual modules. This training-free initialization requires no external code engine; disaster-specific E2A capability is learned from verified traces through E2A supervised fine-tuning, Grounding reinforcement learning, and Action reinforcement learning. The checkpoints and λ are specified in the experimental setup.

E2A SFT: Structured-Trace Cold Start

Task-vector merging provides complementary priors; E2A fine-tuning turns them into links among disaster objects, computational states, documented rules, and outputs. We fine-tune Disaster-Codex on the verified traces produced by DisasterSAGE-Agent. Given an image and task question, the model generates the six E2A fields followed by the deterministic target, serialized as one reasoning block and one answer block.

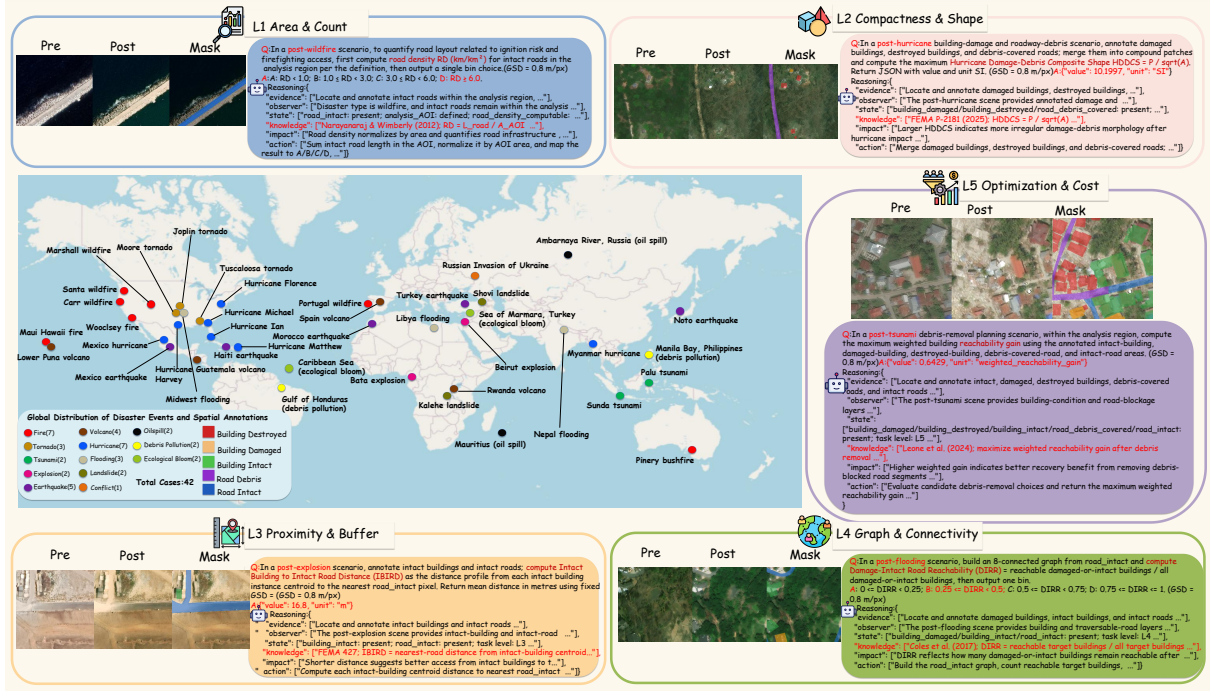


Figure 3: Representative E2A samples from mask-complete evidence to categorical, binary, and quantitative targets across five levels.

This stage establishes the output protocol and initializes a policy that can bind visible objects to subsequent variables and operations. The resulting imitation-based cold start is then strengthened by two reward-driven stages that directly optimize evidence-set completeness and task-specific numerical tolerance.

Progressive E2A Reinforcement Learning

After the initialization and E2A SFT stages in Figure 4, we use Group Relative Policy Optimization (GRPO) (Shao et al. 2024): Grounding RL optimizes task-relevant evidence selection, and Action RL further optimizes rule-constrained decision generation. This ordering follows the dependency structure of E2A reasoning. Area, buffer, graph, and optimization operators operate on recovered objects; therefore, missing or duplicated evidence alters the state representation used for action reward evaluation.

Stage 1: Grounding RL Trace decomposition yields 119,134 localization instances. The predicted box set P is matched to the non-empty reference box set G . Greedy matching by intersection over union (IoU) returns π^* , a set of matched pairs (p, q) with $p \in P$ and $q \in G$. The format score f_G averages four checks: reasoning closure, the answer marker, structured box output, and valid normalized coordinates. The final grounding reward r_G combines mean matched IoU r_{IoU} , count consistency r_{cnt} , and f_G :

$$\begin{aligned} r_{\text{IoU}} &= \frac{1}{|G|} \sum_{(p,q) \in \pi^*} \text{IoU}(p, q), \\ r_{\text{cnt}} &= \frac{\min\{|P|, |G|\}}{\max\{|P|, |G|\}}, \\ r_G &= .72r_{\text{IoU}} + .18r_{\text{cnt}} + .10f_G. \end{aligned} \quad (3)$$

Eq. (3) implements the 0.8/0.2 IoU-count and 0.9/0.1 accuracy-format mixtures; fallback coordinate lists halve f_G .

Stage 2: Action RL Action RL optimizes one result per task instance. For a scalar quantitative prediction $\hat{y}$ and target y , the normalized error is $\epsilon = |\hat{y} - y| / \max(|y|, 1)$. Eq. (4) defines its tolerance reward $w(\epsilon)$:

$$w(\epsilon) = \begin{cases} 1, & \epsilon \leq .02, \\ (.50 - \epsilon) / .48, & .02 < \epsilon < .50, \\ 0, & \epsilon \geq .50. \end{cases} \quad (4)$$

The task type t uses C for categorical, B for binary, Q for quantitative, and Q_u for unit-bearing quantitative. For categorical tasks, c is the exact option-identifier match and $\rho \in [0, 1]$ is the normalized SequenceMatcher ratio between predicted and target option text; only exact option text gives $\rho = 1$. For binary tasks, b is the exact decision match. For unit-bearing quantitative tasks, $\hat{u}$ and u are the predicted and target units. With $\mathbf{1}[\cdot]$ denoting the indicator function, Eq. (5) defines $r_{\text{ans}}(t)$:

$$r_{\text{ans}}(t) = \begin{cases} .5c(1 + \rho), & t = C, \\ b, & t = B, \\ w(\epsilon), & t = Q, \\ .9w(\epsilon) + .1 \times \mathbf{1}[\hat{u} = u], & t = Q_u. \end{cases} \quad (5)$$

The format score f_A checks reasoning closure, the answer marker, and answer validity. The total reward combines answer and format credit as $r_A = .9r_{\text{ans}} + .1f_A$. The denominator floor stabilizes training for small-magnitude targets, whereas test scoring follows Eq. (6). Both stages use group-normalized rewards with Kullback-Leibler regularization.

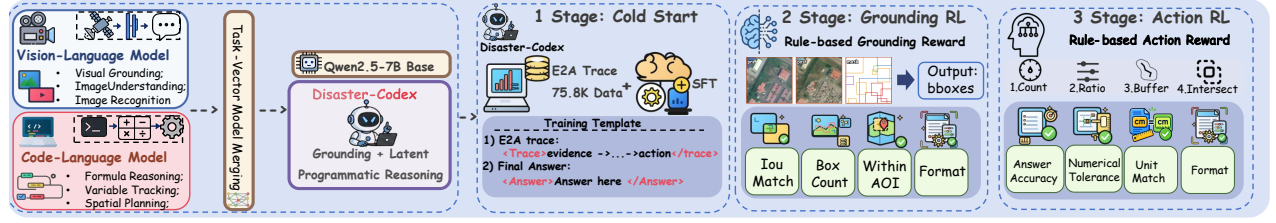


Figure 4: Three-stage Disaster-Codex training: task-vector initialization, E2A SFT, Grounding RL, and Action RL with rule-based rewards across five reasoning levels.

Backbone VLM	Disaster Types (Score $\uparrow$)													Avg.
	Land.	Earthq.	Torn.	Conf.	Fire	Expl.	Tsun.	Hurr.	Volc.	Flood	Oil	Debris	Bloom	
<i>Zero-shot Open VLMs (Base Size)</i>														
Qwen2.5-VL-7B	0.69	36.31	12.57	9.33	6.23	2.69	4.14	28.04	49.80	29.07	25.00	6.52	21.43	17.83
LLaVA-OV-7B	0.00	47.41	6.59	9.84	17.45	19.39	3.31	26.85	47.44	30.45	8.33	10.87	14.29	18.63
MiniCPM-V-8B	0.69	37.46	7.49	51.30	25.98	5.39	3.64	32.94	49.21	30.20	11.11	17.39	7.14	21.53
InternVL3.5-8B	1.39	1.59	1.05	0.00	15.12	1.26	1.49	2.15	2.17	1.38	13.89	2.17	21.43	5.01
Qwen3-VL-8B	0.69	47.55	25.15	8.81	11.30	21.01	11.92	26.49	55.51	28.57	36.11	17.39	0.00	22.35
Gemma-3-12B-it	0.00	30.84	4.34	8.29	18.50	13.64	1.16	28.04	48.82	31.20	69.44	13.04	7.14	21.11
<i>Zero-shot Open VLMs (Large Size)</i>														
Ovis2-16B	0.00	26.08	12.72	7.25	11.54	8.26	8.28	38.66	41.54	48.75	72.22	21.74	21.43	24.50
Mistral-Small-24B	0.69	41.50	8.08	11.92	19.58	8.26	5.79	24.82	48.03	26.57	25.00	28.26	57.14	23.51
Qwen3-VL-30B-A3B	0.69	38.33	8.83	17.10	11.30	17.77	7.12	30.19	50.59	36.59	41.67	13.04	21.43	22.67
<i>Zero-shot Open VLMs (Extra-Large Size)</i>														
Qwen2.5-VL-32B	0.69	41.50	12.57	25.91	8.93	6.46	9.93	24.58	51.18	27.32	52.78	10.87	21.43	22.63
Qwen3-VL-32B	0.69	47.55	19.46	36.79	28.27	7.90	7.95	31.15	58.46	40.73	50.00	13.04	0.00	26.31
Qwen2.5-VL-72B	0.69	44.96	22.75	9.33	11.94	5.57	11.42	26.61	50.79	28.32	52.78	13.04	0.00	21.40
<i>In-domain Disaster-Codex Checkpoints</i>														
Disaster-Codex (7B)	0.69	36.17	10.93	8.81	5.63	2.87	6.13	26.37	50.00	28.07	41.67	30.43	7.14	19.61
+ E2A SFT	0.69	60.09	19.01	79.27	55.81	36.80	12.75	66.11	49.02	47.49	5.56	36.96	42.86	39.42
+ Grounding RL	0.69	49.86	27.84	80.83	63.97	36.45	13.58	66.71	58.86	47.24	61.11	15.22	64.29	45.13
+ E2A Action RL	40.97	76.51	52.25	83.42	82.11	52.42	30.30	73.51	65.55	72.93	69.44	89.13	92.86	67.80

Table 3: Macro scores by disaster type on the fixed test split. Zero-shot VLMs are grouped by model size; Disaster-Codex rows show successive alignment stages.

Experiments

Experimental Setup

Design. We evaluate disaster-level coverage, 5-level stage progression, matched trace supervision, compute scaling, and residual errors on the fixed 7,587-sample test set. The zero-shot pool covers Qwen2.5-VL (Bai et al. 2025b), Qwen3-VL (Bai et al. 2025a), LLaVA-OneVision (Li et al. 2024a), and MiniCPM-V (Yao et al. 2024). It also includes InternVL3.5 (Wang et al. 2025b), Gemma 3 (Gemma Team et al. 2025), Mistral-Small (MistralAI 2025), and Ovis (Lu et al. 2024). All zero-shot VLMs receive the same inputs and instructions; every Disaster-Codex checkpoint uses identical test identifiers.

Metrics. Sample n has answer type t_n , with C for categorical, B for binary, Q for quantitative, and Q_u for unit-bearing quantitative. Categorical and binary outputs are normalized and matched exactly. For quantitative component j , prediction $\hat{y}_{nj}$, target y_{nj} , and component error δ_{nj} are linked by $\delta_{nj} = |\hat{y}_{nj} - y_{nj}|$ when $|y_{nj}| < 10^{-6}$, and $\delta_{nj} = |\hat{y}_{nj} - y_{nj}|/|y_{nj}|$ otherwise. For categorical or binary samples, $\hat{y}_n$ is the normalized prediction and y_n is the normalized target. For quantitative samples, J is the num-

ber of components; for unit-bearing samples, $\hat{u}_n$ and u_n are canonicalized predicted and target unit strings. The sample score σ_n is

$$\sigma_n = \begin{cases} \mathbf{1}[\hat{y}_n = y_n], & t_n \in \{C, B\}, \\ \prod_{j=1}^J \mathbf{1}[\delta_{nj} \leq 0.2], & t_n = Q, \\ \mathbf{1}[\hat{u}_n = u_n] \prod_{j=1}^J \mathbf{1}[\delta_{nj} \leq 0.2], & t_n = Q_u. \end{cases} \quad (6)$$

Thus near-zero targets use an absolute tolerance of 0.2, whereas others use a 20% relative tolerance. All vector components must pass, and unit-bearing answers must exactly match the target unit. The soft unit term provides graded training feedback, while evaluation requires exact agreement. These tolerances are deterministic acceptance rules rather than estimates of calibration, operational uncertainty, or measurement precision. Categorical answers are scored by option identifier, with identifier-text agreement logged separately. For disaster-type and E2A-level evaluation, we report 100 times the unweighted mean of group-wise sample averages, preventing frequent groups from dominating.

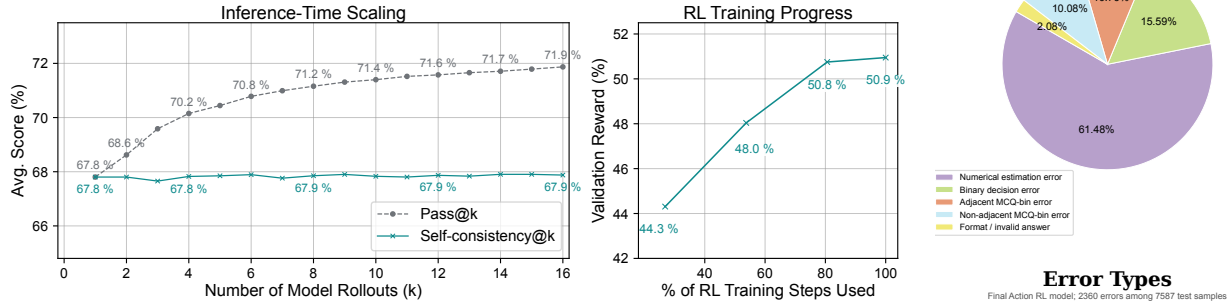


Figure 5: Inference scaling (left), Grounding-RL progress (center), and residual Action-RL errors (right).

Implementation. We construct Disaster-Codex by merging Qwen2.5-VL-7B-Instruct (Bai et al. 2025b) and Qwen2.5-Coder-7B (Hui et al. 2024) around their shared Qwen2.5-7B language backbone with $\lambda = 0.95$. Only language parameters are merged; the vision encoder, projector, processor, and visual-token rows remain from the VLM. SFT uses LLaMA-Factory (Zheng et al. 2024) with rank-16 LoRA ($\alpha = 32$, dropout 0.05) on language projections while vision remains frozen. Three bfloat16 epochs on eight NVIDIA H100 80 GB GPUs use 8,192-token sequences, AdamW, and a peak learning rate of 10^{-4} , taking 11.77 h (94.16 GPU-hours). RL uses EasyR1 on HybridFlow/veRL (Sheng et al. 2025) with FSDP, four rollouts at temperature 1, global batch size 32, learning rate 10^{-6} , and KL coefficient 10^{-2} . On four H100 GPUs, Grounding RL runs 3,722 steps for 99.68 h (398.70 GPU-hours; maximum 4,096 tokens), whereas Action RL runs 1,895 steps for 31.05 h (124.18 GPU-hours; maximum 1,024 tokens). Total training cost is 617.04 H100 GPU-hours.

Main Results

Performance across disaster types. Table 3 reports macro scores over 13 categories. Successive E2A SFT and RL raise Disaster-Codex from 19.61 to 67.80, while the strongest zero-shot reference reaches 26.31. Action RL improves all 13 categories, showing that complete evidence recovery and rule-constrained action learning transfer across terrestrial and marine disasters.

Model	L1	L2	L3	L4	L5	Avg.
Disaster-Codex	17.69	3.51	22.09	10.68	0.00	10.79
+ E2A SFT	48.97	8.63	32.52	69.15	4.00	32.65
+ Grounding RL	56.57	8.31	33.09	69.47	40.00	41.49
+ Action RL	73.65	67.41	48.84	86.14	56.00	66.41

Table 4: Full-data stage progression across E2A levels.

Performance across E2A levels. Table 4 shows that full-data training raises the level-macro score from 10.79 to 66.41. SFT mainly improves L1 and L4, Grounding RL boosts L5, and Action RL improves all levels. Because each

Supervision Source	L1	L2	L3	L4	L5	Avg.
RS-EoT	20.65	0.32	18.77	12.03	0.00	10.35
VLM Distillation	42.65	22.68	31.03	60.79	48.00	41.03
DisasterSAGE E2A	64.01	26.20	38.19	69.74	48.00	49.23

Table 5: Matched-2K supervision-source ablation.

checkpoint inherits its predecessor, the results reflect a curriculum rather than isolated stage effects. The strong L2 and L5 gains suggest that rule alignment is particularly useful for geometric reasoning and optimization.

Trace, Scaling, and Error Analysis

Trace supervision. Under matched model, data, and optimization settings, Table 5 shows that DisasterSAGE E2A achieves 49.23, outperforming VLM distillation at 41.03 and RS-EoT (Shao et al. 2026) at 10.35. Although individual trace fields are not isolated, the 8.20-point gain over VLM distillation supports structured E2A supervision over generic rationale transfer.

Compute scaling and residual errors. Figure 5 shows that larger rollout budgets improve Pass@ k but not self-consistency, indicating that majority voting misses some correct candidates. Grounding-RL validation reward increases steadily. Numerical estimation accounts for 61.48% of residual errors, with the remainder arising from binary decisions, categorical bins, and formatting.

Conclusion

We introduce DisasterSAGE for auditable disaster reasoning from remote sensing imagery. DisasterSAGE-Agent converts complete evidence, expert knowledge, and deterministic execution into verifiable E2A traces, while DisasterSAGE-Bench structures these traces as multi-hazard executable tasks. To support effective learning of E2A reasoning, we further introduce Disaster-Codex, which focuses on evidence recovery and rule-constrained action generation. Collectively, our work supports verifiable evidence-to-action reasoning for reproducible disaster assessment.

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